Tactix certified UNSAT verticals
Public Direct PHP and live GF(3)/Tseitin foundations

Development checkpoint for Andrew

4 October 2026

Outcome

Three independent theory-to-proof increments have landed. Direct PHP now completes one public terminal controller vertical. GF(3) Gaussian elimination and Tseitin exact-XOR reasoning now reach authenticated live publication through private adapters. This is an implementation checkpoint, not benchmark acceptance: no target, WV/WV2, Tier-D, or regression campaign was run, and the saved `d170b4a5` comparison was not rerun.

Mechanism	Implemented boundary	Status
Direct PHP	Live recognition, exact replay, DSR/DRAT/no-proof terminal refutation, public registry/C/online/random execution, raw feedback	Public, default disabled
GF(3)	Exact live owner, chained Source/Scale/Add elimination, independent RUP validation, atomic unit/conflict publication and derivation journal	Private
Tseitin	Complete live capture, exact width-2–4 XOR discovery, cautious graph admission, independently replayed X7/DSR unit publication	Private

Direct PHP public action

Commit `e4d76b336` appends `DirectPhpRefutation` at registry ID 36, disabled StepId 25, and C parameter tag 39. One canonical request carries exact snapshot, recognition, certificate, and replay limits. CLI and C API solvers own a Kernel-bound noncopyable session. Explicit online programs and the random controller execute the same serialized action; DefaultPolicy excludes it.

The action recognizes a complete direct $(H+1)$-pigeon/H-hole encoding and publishes the recursive RAT/RUP certificate atomically. Success returns terminal UNSAT, so validation rejects any propagation follow-up. Feedback separates terminal status, capture/recognition/replay work, and proof additions, deletions, literal volume, helpers, and maximum width.

The CLI now prints action attempts separately from applications and decline classes. The fuzz reachability ledger keys `DirectPhpRefutation` on this attempt counter, so a future random-policy leg can require the public execution boundary to fire rather than mistaking registration for coverage. The effort ledger classifies the action as typed, public, default-disabled, and explicitly governed.

A real parser boundary issue was found during integration. Non-LRAT input finalization intentionally releases binary clause IDs. The adapter now assigns explicitly tagged deterministic capture-local replay identities. It clears those tokens at the proof boundary, where DSR/DRAT deletion is literal-based; they

can never be interpreted as LRAT references. The C API test exercises this parser-finalized DSR path end to end.

GF(3) live Gaussian publication

Commit `86d142984` wraps the authenticated Gaussian bridge and clausal RUP certificate in a cautious live transaction. It supports general chained `Source`/`Scale`/`Add` derivations, rather than only pairwise elimination of original equations. Immediately before commit, the adapter rebuilds and exact-compares the algebraic certificate and physically revalidates every source occurrence, clause, and proof identity.

Only derived units or the empty conflict enter search. Intermediate case-split clauses are proof-only. DSR, DRAT, and no-proof modes are supported; LRAT/PBP and proof-backed split publication decline. The bounded persistent journal maps each derived row and right-hand side to its original GF(3) source combination, Boolean premises, and proof events. It introduces no Boolean variable and no PB step. Nonunit reduced rows remain unsupported.

The focused chained fixture uses $x + y = 0$, $y + z = 0$, and $x + z = 1$. No original pair produces a unit; chained elimination derives $x = 2$, $y = 1$, and $z = 2$, and the test independently RUP-checks the resulting Boolean units. A separate inconsistent system publishes the empty clause and returns UNSAT.

Tseitin autonomous discovery

Commit `048442c40` adds a bounded frontend over the existing live X7 adapter. It captures the complete live binary and large-clause population, derives only complete width-2–4 XOR groups, and rejects opposite parities, incomplete encodings, bad edge degree, even charge, residual edge touches, stale state, interruption, and exhausted limits. Stable group identity and traversal are deterministic. The downstream adapter does not trust the plan: it recaptures and independently authenticates it before X7/DSR publication.

The first version accepts one connected XOR population. It declines multiple disconnected populations rather than silently choosing a component. Public integration must expose component selection, exact replay, feedback, and scheduling as controller-visible data.

Evidence and limits

Each increment passed focused Debug and ASan/UBSan validation. Fresh combined assert-enabled `-O1` and ASan/UBSan builds each passed 17/17 focused tests: Direct PHP action/live/public/API/random paths; GF(3) bridge/certificate/live paths; Tseitin admission/discovery/live/plan/transaction; and shared scheduling, serialization, API-instance, and inprocessing checks. Sanitizer execution used `detect_leaks=0` because LSAN cannot run in this environment. The documentation/configuration suite also passed, including the action inventory, feedback, registry/policy, effort-governor, and fuzz-reach conformance checks.

The combined run exposed two stale tests from the earlier affine lifecycle repair. One invoked Lucky-8 outside its restored structured-root position; the other still expected three structural stages rather than the new affine stage plus three structural stages. Commit `5c23c1f4c` updates the expectations and both pass. This was test debt, not a solver failure.

No instance recovery or speedup follows from these unit tests. Direct PHP still needs effective both-policy fuzz, full Debug, target measurement, and the four regression gates. GF(3) needs a public typed outcome action with replay and raw feedback. Tseitin needs a public component payload and session. All intended enabled behavior still owes N1–N4, B28+C5, Rank1–50, WV100, and WV2–89 with the 1,800-second proof-check cap and the documented completed-proof slowdown allowance.

Next actions

When benchmark execution is authorized, first measure `cabp`, Break, and Timetable on the repaired affine scheduler before launching broad gates. In parallel, expose GF(3) Gaussian outcomes and Tseitin connected components as typed learner-selectable actions without embedding hidden opportunity policy. Use those actions on the fixed-44 and LymphoSAT-owned WV/WV2 rows; failures should drive premise, consequence, variable, and proof-cost analysis rather than restoring a sequential portfolio or a dedicated UNSAT solver.

The detailed technical record is `benchmarks/reports/2026-10-04-certified-unsat-verticals.md`.